RED ZONES IN THE HIMALAYA

These are regions where rising Himalayan temperatures, unstable glacial lakes and transboundary rivers threaten cascading disasters





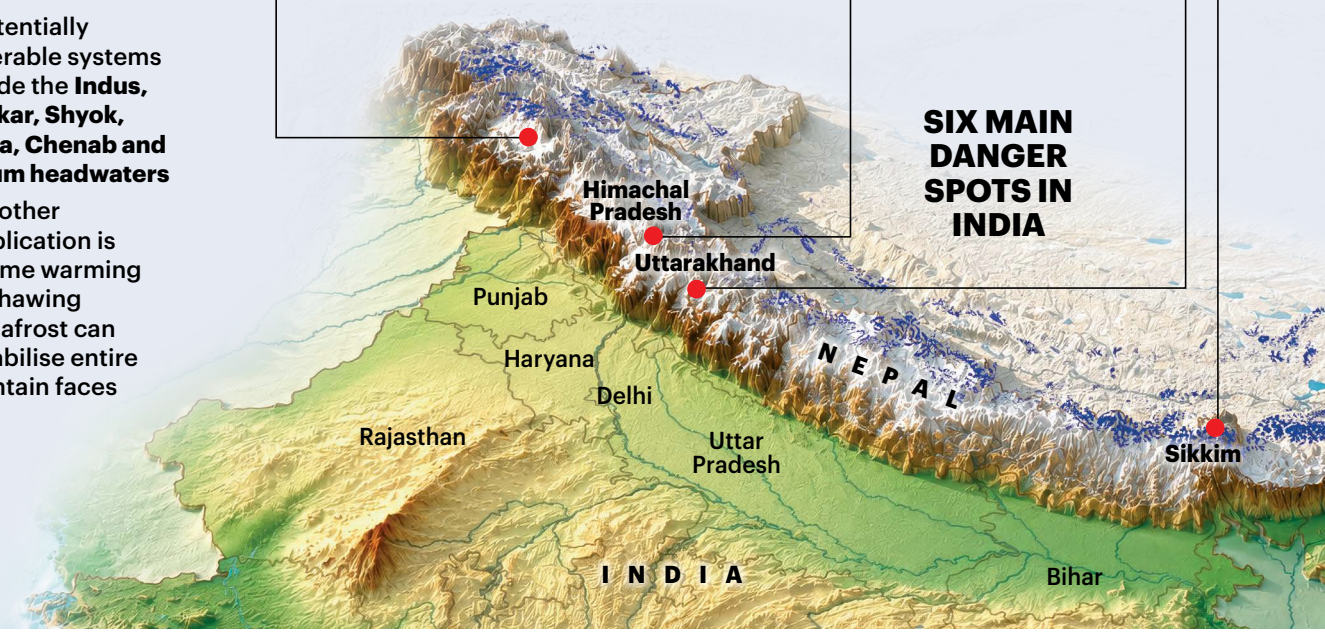
HIMACHAL PRADESH

- The **upper Sutlej basin**, including tributaries coming from Tibet, is heavily exposed
- Himachal's **Ghepang Ghat glacial lake** in particular illustrates how fast some of these lakes are changing. ISRO found its area increased from about **36.5 hectares in 1989 to 101.3 ha in 2022—a 178 per cent expansion**
- Baspa, Spiti, Parvati and upper Sutlej catchments are seen as vulnerable zones

LADAKH AND JAMMU & KASHMIR

- Potentially vulnerable systems include the **Indus, Zanskar, Shyok, Nubra, Chenab and Jhelum headwaters**
- Another complication is extreme warming and thawing permafrost can destabilise entire mountain faces

SIX MAIN DANGER SPOTS IN INDIA



Lhonak Lake and helped generate a surge that breached its moraine dam. About 50 million cubic metres of water, equivalent to nearly a third of Delhi's water needs in a month, rushed into the Teesta system. The flood destroyed the 1,200 MW Teesta III dam at Chungthang, swept away bridges, roads, military installations and homes, killed scores and affected more than 88,000 people. It demonstrated how a natural hazard can become a cascading infrastructure disaster when dams and settlements occupy the same narrow river corridor. Other expanding lakes above the Teesta keep Sikkim—and northern West Bengal downstream—in the front rank of risk.

On the western Himalayas, Uttarakhand may now have the most dangerous combination of unstable high mountains and dense human exposure. Its Ganga and Yamuna headwaters are crowded with pilgrimage towns, hotels, roads, bridges, tunnels and hydropower works. The Alaknanda-Dhauliganga, Bhagirathi, Mandakini and Kali-Sharda systems are particularly vulnerable. In 2013, extreme rainfall and a burst of water from Chorabari Tal created the Kedarnath disaster. In 2021, a rock-and-ice avalanche from Ronti Peak near Chamoli sent about 27 million cubic metres of material down the Rishiganga-Dhauliganga, destroying one hydropower project, damaging another and killing or leaving missing about 200 people. There had been no large glacial lake waiting to burst.

The warning has since sharpened. An April 2026 study by Ashim Sattar of IIT Bhubaneswar and Nandu Krishnan of the Divecha Centre for Climate Change in IISc Bengaluru identified 219 hanging glaciers in Uttarakhand's Alaknanda basin alone and modelled potentially enormous avalanche flows in the Badrinath-Mana sector. Here, relatively modest official counts of high-risk lakes conceal the larger threat from suspended ice, unstable rock and extreme development along valley floors. Nearby Himachal Pradesh's principal danger lies in the upper Sutlej and its Baspa, Spiti and Parvati